

Module 2 Assignment – The World in 2045



NANYANG
TECHNOLOGICAL
UNIVERSITY
SINGAPORE

Centre for Professional and
Continuing Education



SkillsFuture Career Transition Programme (SCTP)

Advanced Professional Certificate in
Data Science and AI

Get the Skills, Tools and Mindset to
Kickstart Your Tech Career

It is now 85 seconds to midnight

2026 Doomsday Clock Statement

Science and Security Board
Bulletin of the Atomic Scientists

Editor, John Mecklin

January 27, 2026



Founded in 1945 by Albert Einstein, J. Robert Oppenheimer, and University of Chicago scientists who helped develop the first atomic weapons in the Manhattan Project, the Bulletin of the Atomic Scientists created the Doomsday Clock two years later, using the imagery of apocalypse (midnight) and the contemporary idiom of nuclear explosion (countdown to zero) to convey threats to humanity and the planet. The Doomsday Clock is set every year by the Bulletin's Science and Security Board in consultation with its Board of Sponsors, which includes eight Nobel laureates. The Clock has become a universally recognized indicator of the world's vulnerability to global catastrophe caused by man-made technologies.



The Science and Security Board

The *Bulletin's* Science and Security Board (SASB) is composed of a select group of globally recognized leaders with a specific focus on nuclear risk, climate change, and disruptive technologies. [Learn more...](#)

Source: <https://thebulletin.org/doomsday-clock/2026-statement/>

GLOBAL UPDATE: Q1 2026

WORLDWIDE EVENTS

JANUARY

FEBRUARY

MARCH

GEOPOLITICS & ECONOMY



WEF SURVEY:
CONFRONTATION
THREAT

IMF GROWTH
FORECAST:
3.3%



US STRIKES IN
SYRIA (JAN 10)



ICE PROTESTS
(US) - RENEE GOOD



WINTER OLYMPICS (FEB 6-22)
& PARALYMPICS (MAR 6-15)

TECHNOLOGY & ANNIVERSARIES



SPACE SUMMIT
(SINGAPORE, FEB 2-3)



UKRAINE INVASION
4TH ANNIVERSARY
(FEB 24)

MARKET TRENDS



GLOBAL GROWTH:
ROARING START

SEMICONDUCTOR
SALES

+3.7% (JAN)

AIR PASSENGER
DEMAND

+3.8% (JAN)



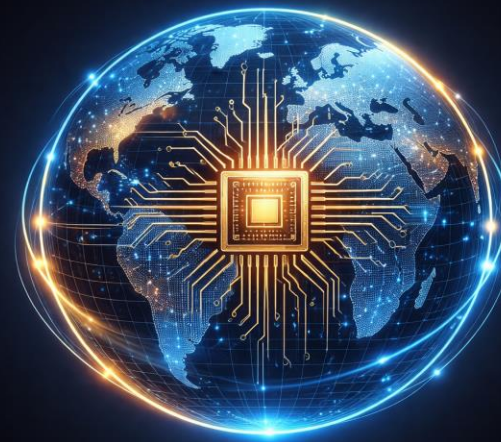
AI IMPACT SUMMIT
(INDIA, FEB 19-20)

Other unresolved conflicts -

1. Venezuela
2. Ethiopia-Eritrea
3. Mali and Burkina Faso
4. DR Congo
5. Mexico
6. Israel-Palestine
7. Afghanistan-Pakistan
8. Cuba *and more...*

Source: <https://www.crisisgroup.org/> as of Mar 7, 2026

Source: <https://share.google/aimode/NkhTew1YkL0kn7BSO> as of Mar 7, 2026



Project Name: The World in 2045



Projected Doomsday Clock in 2045



Doomsday Clock 2045: Tracking Global Existential Risk

Present Day (2024–2025): 85 Seconds to Midnight



85 SECONDS TO MIDNIGHT

Current Status: The closest the clock has ever been to midnight, representing record-level existential risk.

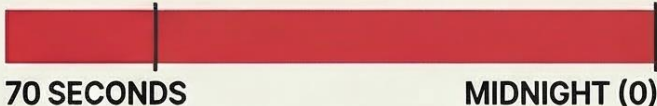


2045 Projection: 70 Seconds to Midnight



70 SECONDS TO MIDNIGHT

A plausible estimate showing a modest deterioration in the global risk profile based on World2045 analysis.



PROJECTED TEMPORAL SHIFT IN RISK:
**Modest Deterioration,
A Fragile but Not Collapsed Order**

Systemic risk remains within the same critical band as today without accelerating dramatically, indicating a persistent high-risk environment.

Drivers vs. Stabilisers

Primary Drivers of Deterioration



Continued climate stress, uneven governance, and persistent regional conflicts push the clock forward.



Strong growth in Europe, Asia, and the Americas prevents a catastrophic acceleration.



Stabilising Development Trajectories

The World2045 Model



An analytical framework integrating indicators like governance stability, climate vulnerability, and adaptation readiness.



Project Name: The World in 2045

GitHub Repository URL

<https://tinyurl.com/world-in-2045>

WORLD2045_BLUEPRINT_ORIGINAL.md
README.md

Business Objectives

- Develop a snapshot of the world in 2045
- Which countries are likely to strengthen, sustain, or struggle in their development trajectory by 2045?

Business Value Proposition

- Relevant parties can plan for actions to encourage positive growth and remediate negative trajectories

Target Audience

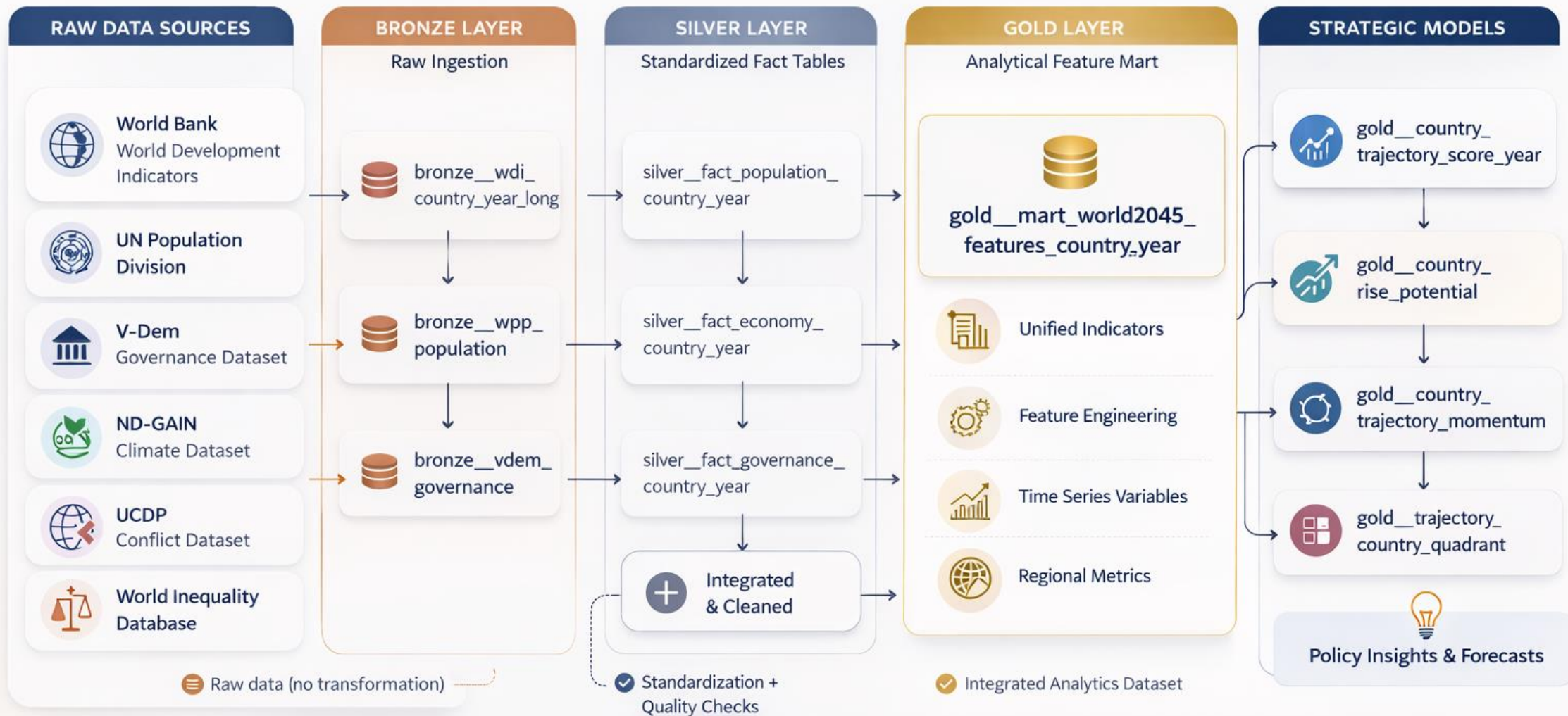
- United Nations
- Key Global Groups - Economic, Political, Social & Civil Society, Geopolitical & Developmental
- Nation-state Governments



World2045

Data Lineage & Analytics Architecture

From Global Data Sources to Country Trajectories and Rise Potential



Key Capabilities



Global Coverage



Country-Level Precision



Temporal Analysis



Comparative Metrics



Predictive Models



Data Transparency

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<https://tinyurl.com/world-in-2045>

TECHNICAL_README.md

Analytical Requirements

- Grain - 'country_iso3' and 'year'
- Analysis - historical trends, comparative country, development trajectory, long-term scenarios
- Forecasting models - output read as structured continuation paths, not as full causal forecasts of politics, climate shocks, or war

Conformance Schema

- Dimension tables shared across multiple fact tables (Kimball dimension modelling)
- Simplified integration (joins) due to structural alignment

Data Indicators.

Population

Life expectancy

GDP & GDP per capita

Governance

Adaptation readiness

Climate vulnerability

Conflict severity

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Technology stack

Component	Technology
Warehouse	Google BigQuery
Ingestion	Python
CI	GitHub Actions
Transformation	dbt
Visualization	Streamlit & Hugging Face
Version Control	GitHub

GitHub Repository URL
<https://tinyurl.com/world-in-2045>
WORLD2045_BLUEPRINT_ORIGINAL.md
README.md

auto-scaling, in-storage computing, fine-grain control
↑ control, flexible, easier debugging, satisfies current need
automation, seamless integration
industry standard, built-in testing
multipage pattern, HF free-tier
reliable, industry standard

Technologies to be implemented in future

- | | | |
|--------------------------------|---------------------------|-----------------------------|
| Meltano | Dagster | Great Expectations |
| • Data updates more frequently | • Upscaling of data cycle | • Increasingly complex data |

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FINAL_ANALYTICAL_FINDINGS.md

Main findings at-a-glance

- Europe remain the strongest overall structural region by 2045
- Asia is the most heterogeneous region, with countries across all trajectory categories
- Africa remain the most concentrated in structural-risk outcomes under the baseline scenario
- Guyana is the most prominent high-momentum outlier in 2045
- Global convergence appears limited; structural hierarchy persists into 2045

View Dashboard – <https://sofian75-world2045-dashboard.hf.space/>

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GitHub Repository URL

<https://tinyurl.com/world-in-2045>

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Limitations

Known caveats:

- Forecast structural variables are carried forward, not fully projected (may introduce bias)
- Some countries have partial forecast component coverage
- All scores are relative to period-year normalization, so slight negative movements are does not imply absolute decline

Indicator weightage:

- Heuristic composite index (expert-designed scoring model based on aggregation of sub-indicators)
- Subjective
- Comparable to real-world conceptual frameworks such as the U.N. Development Program, Global Competitiveness Index (WEF) and Notre Dame Global Adaptation Initiative

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Future work enhancements

1. Scenario modeling to include multiple development scenarios – e.g.
 - economic growth pathways and governance reform trajectories
 - conflict risk escalation scenario
2. Machine learning forecast
3. Monte Carlo Simulation Uncertainty
4. Policy Sensitivity Analysis Future models
5. Expand indicator coverage to incorporate -
 - Infrastructure quality
 - Infrastructure adaptation
 - Energy transition metrics
 - Water stress exposure
 - Technological readiness
 - Institutional capacity
 - Food security vulnerability
 - Ecosystem degradation

Project Name: The World in 2045

Project structure.

GitHub Repository URL

<https://tinyurl.com/world-in-2045>

MAIN

```
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|
|— dbt/
|
|— dashboard
|   | app/
|   | data/
|
|— notebook/ trg
|   | WORLD2045_Training_Manual_Data_Analyst.ipynb
|
|— data/ (small extracts only)
```



Verification against existing online research

Sources



+ Add sources

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Web

Text

YouTube

Search the web for new sources



Web



Fast research



Select all sources



Deep Research report: The Global Strategic H...



Doomsday Clock - Bulletin of the Atomic Scien...



Chat



3. Geopolitical Fragmentation and Nuclear Risk

While the 2026 setting highlights the need to limit nuclear arsenals, the 2045 report describes a world that has largely moved away from international cooperation ¹ ¹¹ :

- **Institutional Decay:** In the "A World Adrift" scenario, the decay of international institutions leads to **"High" conflict risk** ¹¹ .
- **Nuclear Proliferation:** The Middle East is identified as an emerging "Escalation Hotspot" with **"Medium-High" risk for nuclear proliferation** ¹² .
- **Resource Wars:** Scarcity of freshwater and critical minerals is expected to drive "forever wars"—low-intensity but high-duration conflicts that resist traditional resolution ⁵ ¹¹ .

4. Systemic Fragility

The *Bulletin of the Atomic Scientists* notes that the Clock is a metaphor for the perils humanity must address to survive ⁴ . The 2045 research identifies a **"Tragedy and Mobilization"** scenario where climate catastrophe leads to **"Very High" conflict risk** ¹¹ . The fundamental driver of this volatility is the widening gap between the rapid speed of technological change and the slow adaptability of human institutions ⁹ .

In summary, if the Clock is already at **85 seconds to midnight in 2026** due to early AI risks and current climate trends, the **breach of 1.5°C**, the **militarization of AI**, and the **fragmentation of the global order** projected for 2045 would likely push the symbol to its absolute limit ¹



Save to note



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Start typing...

2 sources



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ANY QUESTIONS?



ACKNOWLEDGEMENTS

PROJECT GROUP 5 TEAM MEMBERS

Portions of the technical design, debugging support, and documentation refinement for this project were assisted by ChatGPT (OpenAI).

All final implementation decisions, data engineering work, analytical modeling, and system integration were performed by the project authors.